

Quarterback Play: Progressions, the Pocket, and Off-Script Football

What the quarterback does in the three seconds after the snap, what he does when the three seconds run out, and how to measure the difference between quarterbacks

Third-and-8, fourth quarter. The quarterback takes the snap in the shotgun, drops, and looks hard to his left. Then his head snaps to the middle of the field. He bounces once on his back foot, then again. The two defensive ends have run around his tackles and are closing from behind him, so he takes two quick steps *forward*, into the space they left. Now a defensive tackle shoves a guard backward into his face, and the quarterback spins out to his right and runs toward the sideline. His receivers, who had been running set routes, suddenly do something else: the one who was deep turns and runs back toward him, the one who was short takes off deep, and a receiver from the far side of the field races across. Four and a half seconds after the snap, the ball floats over a cornerback who is still turning around, and the receiver catches it alone near the sideline. The analyst says, “That’s what the great ones do.” What, exactly, did he do? Why did he step *up* and not back? How did the receivers know where to go? And when none of it works, why is throwing the ball into the seats sometimes the best play he can make? This chapter is about the quarterback’s three to four seconds: what he is doing with his eyes, his feet and his judgment, on schedule and off it.

You’ll be able to...

- Narrate a quarterback’s dropback from the snap to the throw: the protection check, the progression on the clock of his feet, and the moment he decides to leave the pocket.
- Count, on a replay, how many reads a quarterback went through before he threw, by following his eyes and his hitches.
- Tell climbing, sliding and escaping the pocket apart, and say why stepping *up* usually beats drifting back.
- Predict where receivers will go when the quarterback scrambles, using the scramble-drill rules.
- Say when a throwaway or a checkdown is the right decision, and why a throwaway must avoid intentional grounding.
- Profile a quarterback from four numbers (time to throw, aDOT, CPOE and EPA per dropback), and explain why the position dominates a team’s value.

You’ll need

- **Dropback Concepts:** two- and three-receiver combinations (smash, dagger, flood and the rest), each with a numbered order of reads. This chapter is about the man who has to work through those numbers.
- **Pass Protection:** the **pocket** as a cup (interior firm, tackles giving ground), the **launch point** where the quarterback sets to throw, **hot routes** when rushers outnumber blockers, and **time to throw** and **pressure** as numbers.
- **Passing Game Fundamentals:** the **drop** as a clock, the **progression** (the order of the quarterback’s looks), **air yards** and **aDOT** (average depth of target).

How to read the diagrams. The conventions of the rest of Part 4: offense blue circles, defense orange squares, *left* and *right* always the offense’s. Q is the quarterback, R the running back, X and Z the outside receivers, H the slot receiver and Y the tight end; the five linemen are the blue circles at the line, with only the center lettered. On defense E is an end, T a tackle, M and W the Mike and Will linebackers, \$ the slot (nickel) cornerback, C a cornerback, FS and SS the safeties. In the moving pictures, faint lines are where a player has been (solid blue for offense, dashed orange for defense); grey dashes are routes as they were *called*; a dashed brown arrow is the ball in flight. Green belongs to the quarterback: his eyes (a dashed green line to the man he is reading), his moves in the pocket, and his **numbered reads**: a white number is a read he hasn’t reached, a green one is where his eyes are now, and a grey one is a read he looked at and passed over. A purple ring marks the defender who decides the read.

The quarterback’s clock

Every dropback is a race against a clock the defense controls, and the quarterback runs a fixed sequence of jobs against it. Coaches teach the sequence by tying each job to a moment in the drop, so that the quarterback’s feet tell him what his eyes should be doing. Here is the order, for an ordinary dropback pass from the shotgun.

1. **At the snap: the protection check.** Before he looks at a single receiver, the quarterback answers one question: did more rushers come than the protection can block? If yes, the ball goes *now* to the hot receiver or a receiver who sight-adjusted his

route, before any progression starts. He already knows, from the protection call and the pre-snap count (Pass Protection), which rusher is his, so this check takes a glance, not a study.

2. **The drop: the first read.** If the rush is covered, his eyes go to the first read as his feet go back. On a well-timed play, the first read is decided as the drop ends and thrown as he sets his feet or on his first hitch.
3. **The hitch: the second and third reads.** If the first read is covered, he takes a short step forward, a **hitch**, which resets his feet so he can throw again, and moves his eyes to read two. Another hitch, read three. Each hitch is a little under half a second.
4. **The last read: the checkdown.** The final look is to a back or tight end a few yards past the line, the safe throw that keeps the play from becoming a loss.
5. **After that: off schedule.** If the ball still isn't out, the play has left its design. The quarterback now has to manage the pocket (step up, slide, or escape), and decide among a scramble, an improvised throw, a throwaway, or taking the sack.

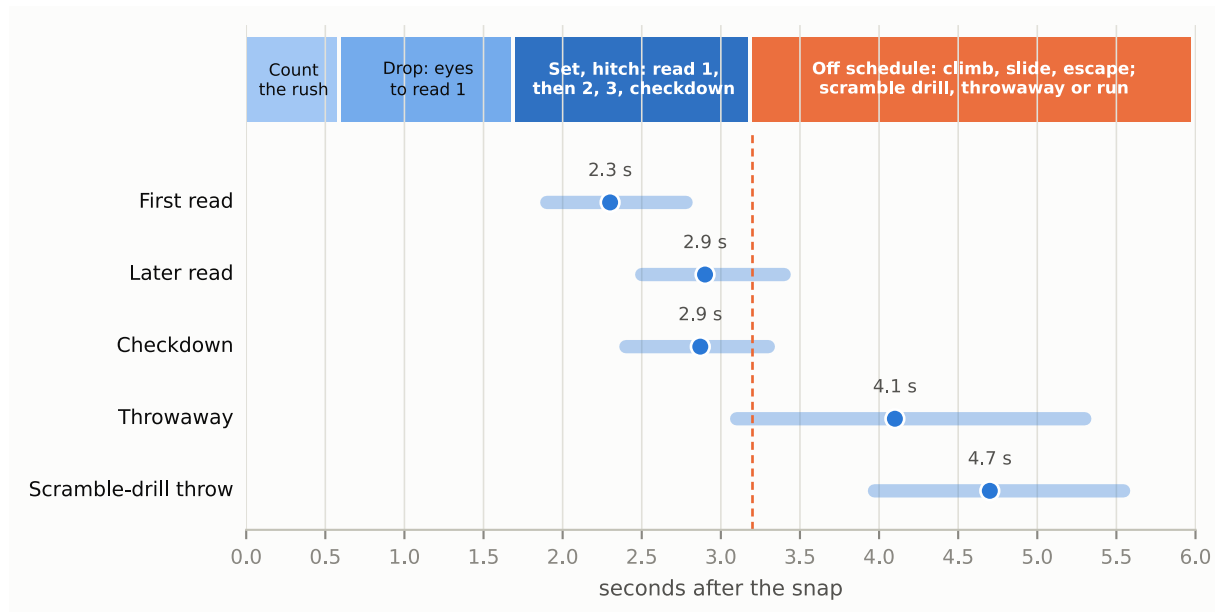


Figure 1: The quarterback's clock. Top: the jobs of an ordinary dropback, tied to the drop and the hitches (typical timings; they vary by play and team). Bottom: when each kind of throw actually leaves his hand in the NFL, from FTN charting of every regular-season pass from 2022 to 2025 that could be matched to a time to throw. The dot is the median, the bar the middle half of throws. First-read throws come out as the quarterback sets or makes his first hitch; later reads and checkdowns cost about another half second; throwaways and scramble-drill throws (throws after the quarterback has left the pocket and the receivers have improvised) live past the three-second mark, off schedule. Data: nflverse play-by-play, FTN charting and participation via nflready.

The bottom half of Figure 1 is the part to remember. The first-read throw leaves the quarterback's hand at a median of 2.3 seconds; a throw to a later read, about 0.6 seconds after that, and the checkdown at about the same moment, because it is the last of the quick looks; and the throwaways and scramble-drill throws at the far right of the chart, after three seconds, when the rush has usually arrived. Every step down the list costs time, and, as you'll see in the checkdown section, value.

! Why it exists

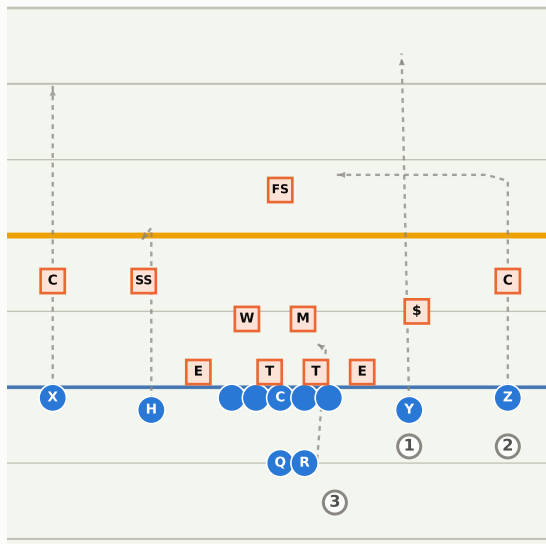
Tying the eyes to the feet solves a problem that sounds trivial and isn't: in the chaos of a rush, how does a quarterback know when to give up on a receiver? Counting seconds doesn't work with three hundred pounds of defensive tackle in his face. Feet do. "Last step, read one; hitch, read two; hitch, checkdown" turns a judgment into a rhythm he can practice ten thousand times, and it keeps him on the same clock as his receivers, whose routes are timed to the same drop (Passing Game Fundamentals).

What it gives up: freedom. A quarterback who throws on rhythm throws to where the play says, not to where a receiver happens to be open. When the defense takes away the design, a rhythm passer can be stuck; that is where the second half of this chapter, the off-script game, comes in.

The progression, live

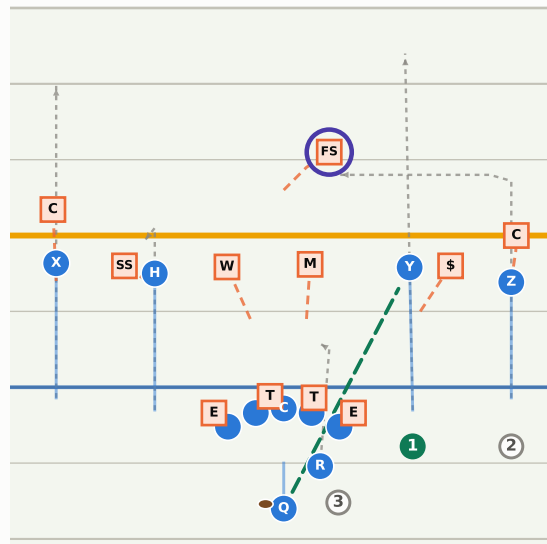
Passing Game Fundamentals drew a progression as numbers on a play diagram. Watched in real time it looks different: the numbers light up one at a time, on the beat of the quarterback's feet, and at each one he is asking a yes-or-no question about one defender, not admiring a receiver.

1. snap



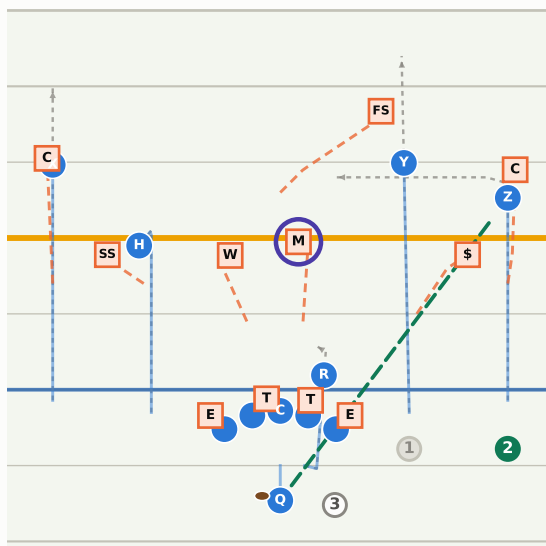
Snap: four rush, four blocked.
No hot throw; eyes go to read 1.

2. +1.2 s



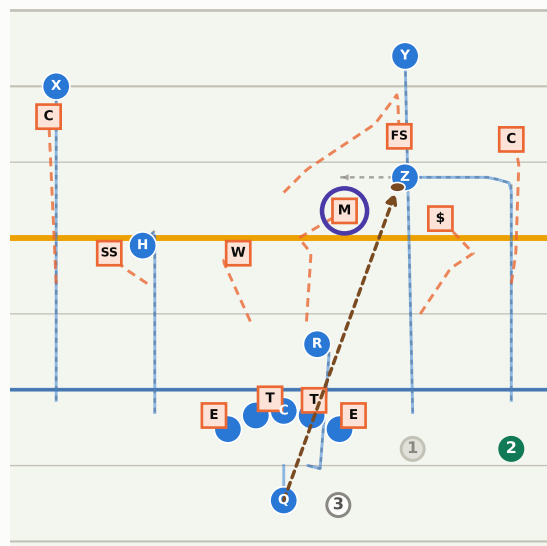
Last step of the drop: read 1, the seam.
The free safety sits on it. No.

3. +2.1 s



Hitch: read 2, the dig, behind the Mike.
Ball out as Z breaks.

4. +3.2 s



Caught in the window. Read 3,
the checkdown, was never needed.

Figure 2: A progression in real time (simulated play). The call is dagger to the right: Y runs up the seam to pull the deep defenders, Z runs a deep dig (an in-breaking route at about 14 yards) behind him, and R is the checkdown. The defense plays a one-high coverage with four rushers. Panel 1: at the snap the quarterback counts the rush (four, all blocked: no hot throw). Panel 2: on the last step of his drop his eyes go to read 1, the seam, and the free safety (ringed) is sitting on it: no. Panel 3: on his hitch he moves to read 2; the dig is about to break in behind the Mike (ringed), and the ball comes out as Z breaks. Panel 4: the ball arrives in the window between the linebackers and the safety. Read 3 was never needed. The grey numbers are reads he looked at and passed over; X's go and H's curl on the left (grey dashes) are not part of this progression.

Three things are happening in Figure 2 that you can't see in a static play diagram.

He reads defenders, not receivers. At read 1 the quarterback isn't asking "is Y open?" He is asking "where is the free safety?" A seam route against a single deep safety is open only if that safety has gone somewhere else, so the safety is the question. At read 2 the question is the Mike linebacker: has he dropped deep enough to take away the dig, or is the window behind him open? Each number in a progression comes with a *key defender*, the man whose position answers it, and good quarterbacks look at him, often never looking at the receiver until the ball is on its way. Which defender is the key on which play depends on the coverage, and that is a subject of its own: Concepts Versus Coverages teaches it, once Part 6 has taught the coverages themselves.

The order is a ranking of risk and reward. Read 1 is the biggest gain the play can produce (a seam is a touchdown if the safety guesses wrong); read 2 a solid one; read 3 a small, safe one. A progression runs from the most valuable throw the defense might allow to the least dangerous one, so that the quarterback takes the big play when it is there and the safe one when nothing else is. Many coaches teach the same thing in three words: *deep to short*.

The ball comes out before the receiver is open. In panel 3 the dig hasn't broken yet; Z is still running straight up the field. The quarterback throws anyway, because he has seen what he needed: the Mike is in front of the window, not in it, and the safety is committed to the seam. That kind of throw has a name.

i Madden vs. real life

In Madden every receiver carries a button icon, the camera sits high behind the quarterback, and you can see the whole field at once and press the button of whoever looks open. A real quarterback sees through a forest of six-and-a-half-foot linemen with helmets bouncing in his face, from eye level, and can usefully look at one area at a time. That is why a progression exists at all: he can't scan everything, so the play tells him what to look at, in what order, and when to give up on each look. When an analyst says a quarterback "got through his progression," he means the quarterback looked at the right things, in the right order, on time, which is much harder than pressing the right button.

Anticipation: throwing to a window that isn't there yet

An **anticipation throw** is a pass released before the receiver is open, usually before he makes his break, aimed at the spot where the window between defenders will be when the ball gets there. It is the bigger, riskier cousin of the timing route of Passing Game Fundamentals: a timing route is thrown to a fixed spot on a fixed step, while an anticipation throw is a judgment, made on any route, about where defenders *will be* in a second.

Why throw to a receiver who isn't open? Because the ball takes time to get there. A firm 20-yard throw is in the air for about a second, and in a second a linebacker can close five yards. A window that is open when the quarterback *sees* it is often closed by the time the ball arrives.

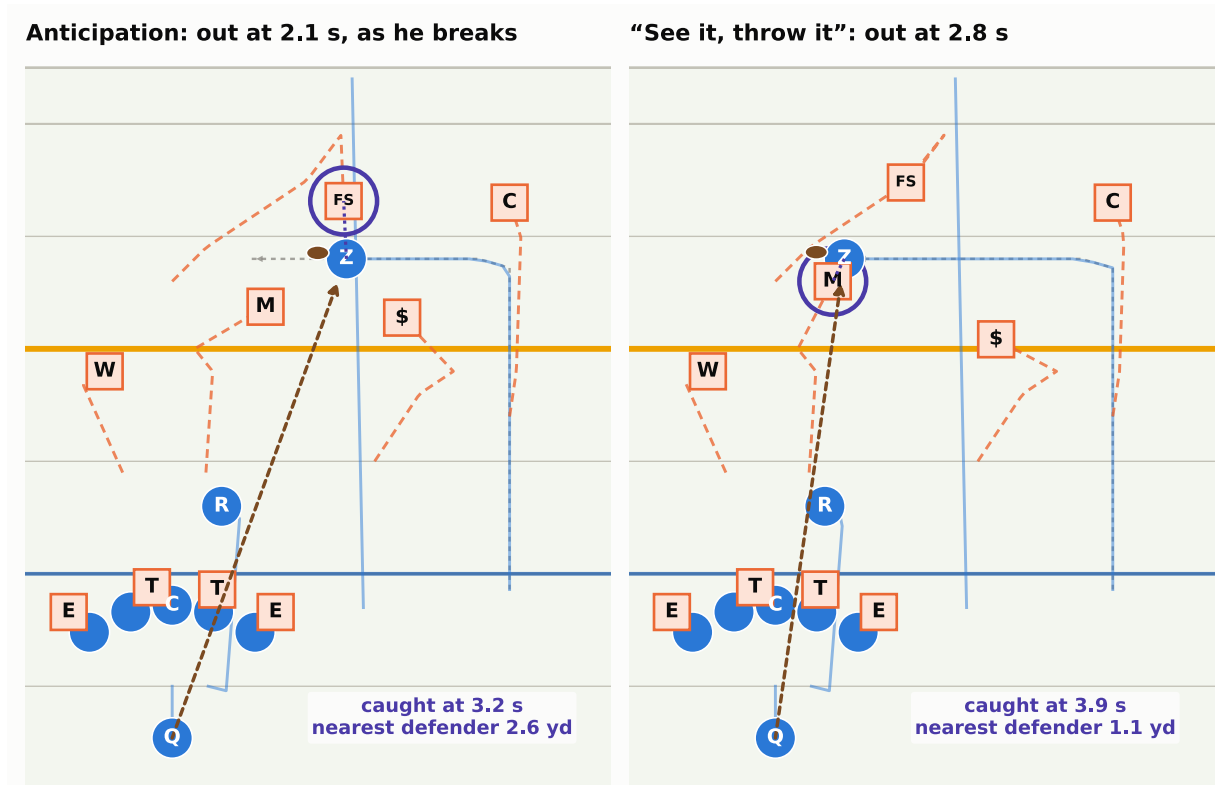


Figure 3: One play, two release times (simulated). Both panels freeze the moment the dig is caught. Left: the anticipation throw, released at 2.1 seconds as Z breaks; the underneath and deep defenders only start toward the ball once it is thrown, and the receiver catches it with room to spare. Right: the quarterback waits until Z looks open and throws at 2.8 seconds. Everything else is the same, but the ball now arrives later and further across the field, where the Mike, reacting to the throw, can close the window. The labels give the distance from the receiver to the nearest defender at the catch.

In the simulation the early throw is caught with the nearest defender 2.6 yards away; the late one with a defender 1.1 yards away, close enough to knock it down or catch it himself. The difference is seven-tenths of a second, about one hitch. That is the whole case for anticipation.

The trade is just as plain. An anticipation throw is a bet on the future, made on information about the present: the quarterback is predicting that the receiver will break where and when he is supposed to, and that the defender he is reading will keep doing what he is doing. If the receiver slips, rounds his cut, or is held up, the ball arrives at an empty spot, often into a defender's hands. And if the defender was disguising his intentions, sitting shallow to bait the throw, the window was never going to open. Anticipation is therefore the skill that most separates veteran quarterbacks from rookies: it takes years of reps with the same receivers to know exactly where they'll be, and years of film to know what a linebacker's first two steps mean.

Coaches often say the young quarterback "needs to see it before he throws it." What they mean is that he is throwing a beat late, waiting for the receiver to look open, which, as Figure 3 shows, is too late against good defenses.

The pocket: climb, slide, escape

The progression assumes a quarterback standing calmly at his launch point. He rarely is. A pocket is a moving thing (the tackles giving ground, the interior bending), and the quarterback has to move inside it without losing his eyes or his feet, so that he can still throw on time from wherever he ends up. Coaches name three moves.

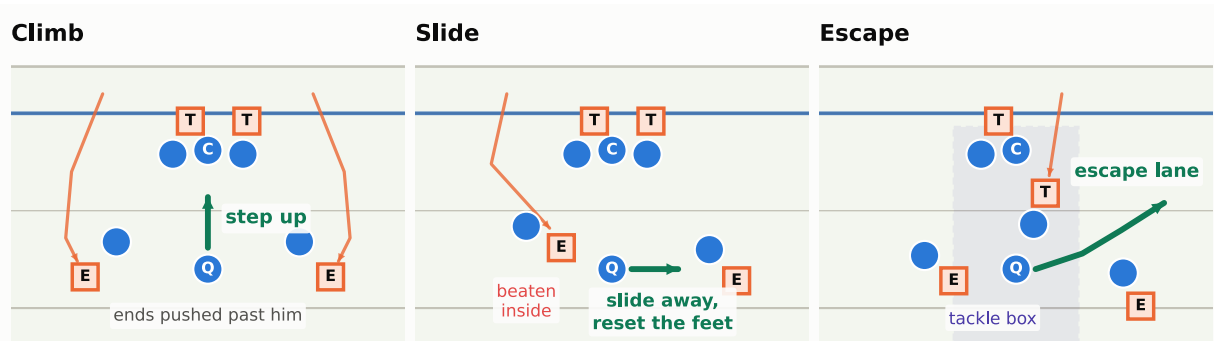


Figure 4: Three pocket moves, each the answer to a different rush picture (a frozen moment about two seconds after the snap; receivers not shown). Left, climb: both ends have run their arcs and the tackles have pushed them deeper than the quarterback, while the interior held, so he steps forward into the cup and the rushers finish behind him. Middle, slide: one end has beaten the left tackle inside, so the quarterback shuffles a couple of yards right, away from the pressure, resets his feet and keeps his eyes downfield. Right, escape: a defensive tackle has driven the right guard back into his face and the left end is coming around, so the pocket is gone; the lane is outside the right tackle, where the other end was run past. Leaving the tackle box (the shaded area between where the tackles lined up) also changes the rules for throwing the ball away.

Climbing the pocket. Climbing the pocket means stepping forward into the pocket, between the interior linemen, as the edge rushers are pushed past. It is the single most important pocket skill, and it follows directly from the geometry of Pass Protection: the tackles give ground on purpose and ride the ends on a wide arc *behind* the launch point, while the guards and center hold firm in front. A quarterback who stays at his launch point, or worse, drifts backward, walks into the spot where the ends' arcs finish. One who steps up two yards lets the ends run past behind him, and throws from the cleanest spot on the field. Quarterback coaches drill the idea endlessly: pressure from the outside is answered by moving *up*, not back.

Sliding. When pressure comes from one side (an end who beat the tackle inside, a blitz through a gap) and the middle is still firm, the quarterback slides a yard or two the other way, keeping his shoulders square to the line of scrimmage and his eyes downfield. A slide buys a half second without ending the play: he is still in the pocket and can still throw to anyone in the progression.

Escaping. When the pocket is gone, usually because the interior collapsed and the quarterback has nowhere to step up into, he leaves. The usual escape lane is outside one of the tackles, on the side where an end rushed too far upfield and was pushed past; it opens exactly because the tackle did his job. Escaping changes the play into something else, which is the next section's subject.

The skill under all three is doing them *without looking at the rush*. A quarterback who drops his eyes to watch the defensive end has stopped reading the coverage, and a quarterback who has stopped reading the coverage has stopped running the play. Great pocket passers feel the rush (from the color at the edge of their vision, from the sound, from the pressure on a lineman's back) and move their feet while their eyes stay downfield. Coaches call this *pocket presence*, and it is one of the few things that clearly separates good quarterbacks from bad ones, as Figure 5 shows in motion.

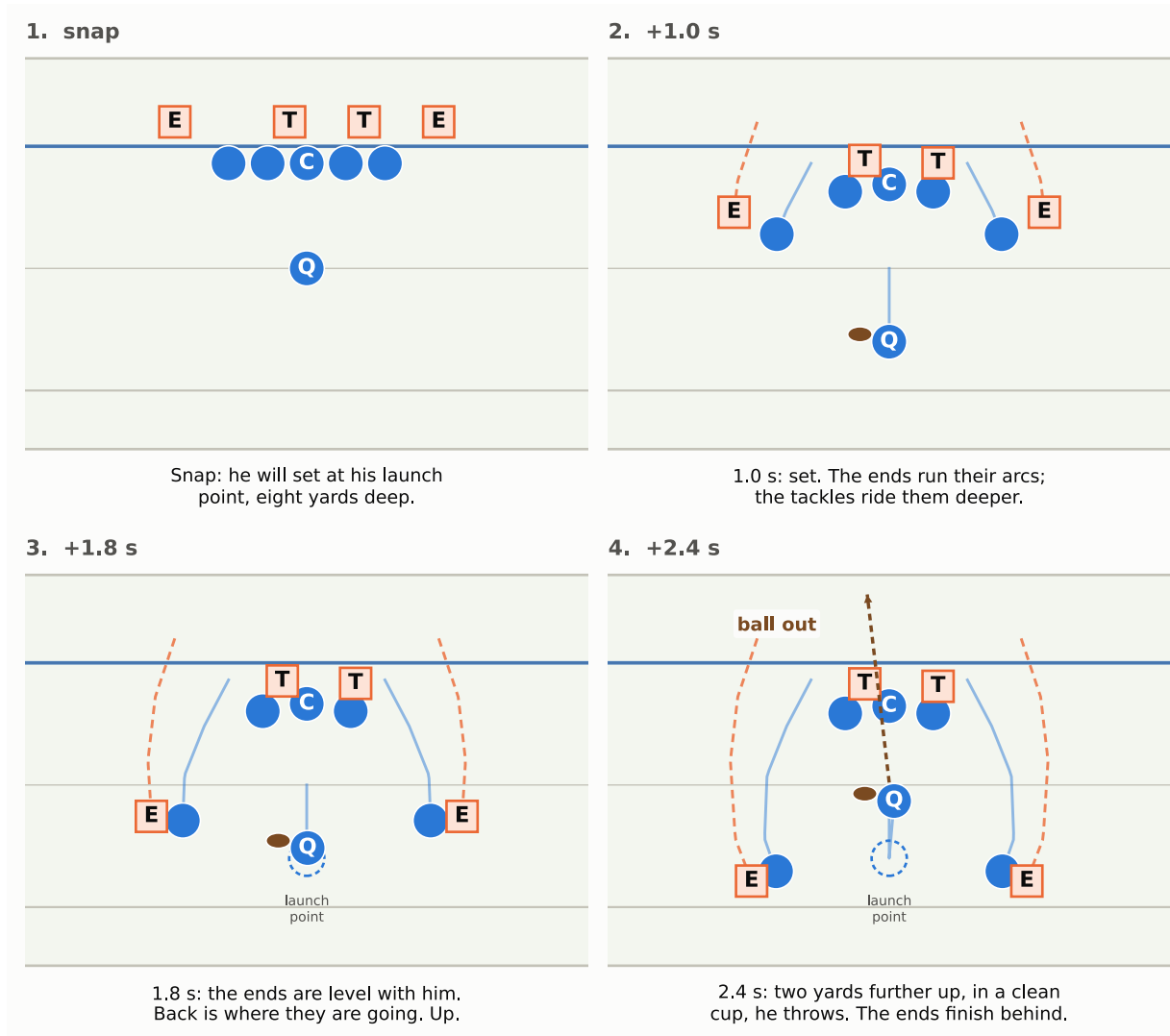


Figure 5: Climbing the pocket, in motion (simulated, the box only). Both ends rush wide around the tackles; the tackles kick back with them and push them deeper than the quarterback’s launch point (the dashed circle), while the guards and center hold their ground against the two tackles. At about 1.8 seconds the ends are level with the quarterback: a drift backward would put him between them. He steps up into the cup instead and throws two yards further upfield, and the ends finish their arcs behind him.

Common misconception

“A mobile quarterback should get out of the pocket whenever it gets crowded.” The quickest way to more sacks is a quarterback who leaves a good pocket. Outside the pocket he has fewer throws (he can see only half the field and has to throw on the run), the rushers who were blocked are now chasing him unblocked, and the sideline is an extra defender. In the FTN charting from 2022 to 2025, quarterbacks were sacked on 8.3% of dropbacks once they were out of the pocket, against 6.2% in it, and their dropbacks out there were worth less on average (+0.011 EPA against +0.049).¹ The best scramblers in the league are good at *not* leaving until the pocket is gone. Then they are devastating, as the next section shows.

Off-script football and the scramble drill

When the quarterback escapes the pocket, or holds the ball past the play’s last read, the play has gone **off script**. An **off-script play** (also called an **off-schedule play**) is one that has left its design: the called routes have run out, the protection has broken, and what happens next depends on improvisation.

¹FTN charting via nflverse ([is_qb_out_of_pocket](#)), merged with nflverse play-by-play; regular-season dropbacks 2022–2025, spikes removed. “Out of the pocket” includes designed rollouts as well as escapes. Author’s calculation via nflready. FTN data dictionary: https://nflready.nflverse.com/articles/dictionary_ftn_charting.html.

Improvisation sounds like the opposite of coaching. It isn't. The quarterback improvises, but his receivers follow rules, practiced every week, that tell them what to do when he starts running. Those rules are the **scramble drill**. Every team has its own version and its own vocabulary, but the core is nearly universal.²

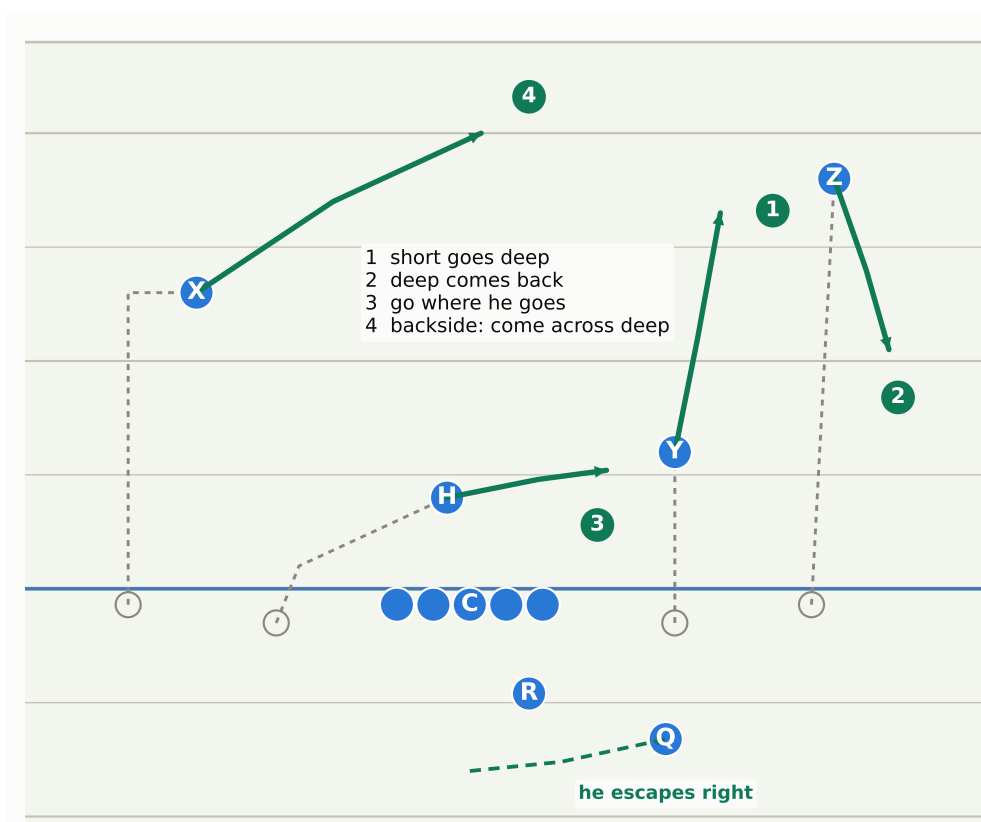


Figure 6: The scramble drill's core rules, drawn at the moment the quarterback escapes to his right (offense only; grey dashes are the routes as called, green arrows what each receiver does once the quarterback leaves the pocket). (1) Short goes deep: Y, sitting on a short hitch, takes off up the field. (2) Deep comes back: Z, who was running deep down the sideline, turns and works back toward the line, along the sideline on the quarterback's side. (3) Go where he goes: H, on a shallow crossing route, keeps crossing toward the scramble side, underneath. (4) Backside receivers come across deep: X, on the far side, works back across the field behind the coverage. The result is four receivers at four depths, all on or toward the quarterback's side and in his line of sight.

Rules differ from team to team; these are the common core.

Each rule has a reason.

- **Short goes deep, deep comes back.** Receivers spread out vertically so the quarterback has a choice of depths, and so that two receivers never end up in the same spot dragging their defenders with them. The deep receiver comes *back* because a quarterback throwing on the run cannot make a 40-yard throw accurately; the short receiver goes deep because the defenders near him have all turned to chase the quarterback, and the space behind them is empty.
- **Go where he goes.** A quarterback running to his right can throw easily to his right and downfield, and only with difficulty back across his body to the left. (Throwing across your body on the run is one of the most dangerous throws in football, because the ball floats back toward the middle of the field where the defenders are.) So receivers work toward his side.
- **Backside receivers come across, deep.** A receiver on the far side can't stand still and wait; he runs back across the field behind the coverage, where defenders who have turned to chase the quarterback have left the deep middle.
- **Get into his line of sight and keep moving.** A receiver the quarterback can't see is not an option. Receivers find grass on a different level from their teammates and stay in his vision, and if he crosses the line of scrimmage to run, they stop running routes and start blocking.

²The rules here are the common core described in coaching and scouting glossaries, for example Inside the Pylon / Scouting Academy, "ITP Glossary: Scramble Drill" (<https://scoutingacademy.com/itp-glossary-scramble-drill/>), and Win With The Pass, "The Scramble Drill" (<https://winwiththepass.com/?p=3138>): "if you're short go deep, if you're deep come back," get into the quarterback's field of vision. Each team's version differs in its details.

Why does any of this work? Because the defense was built for the play the offense called, and that play is over. Against **man coverage**, defenders are running with their backs to the quarterback; when a receiver stops and reverses, his defender has to stop, turn and find him, and he is a step behind. Against **zone coverage**, defenders have dropped to spots that were chosen for the original routes, and they have to leave those spots to chase receivers who are now somewhere else, opening the areas they were guarding. Either way the defense is improvising too, without the rules.

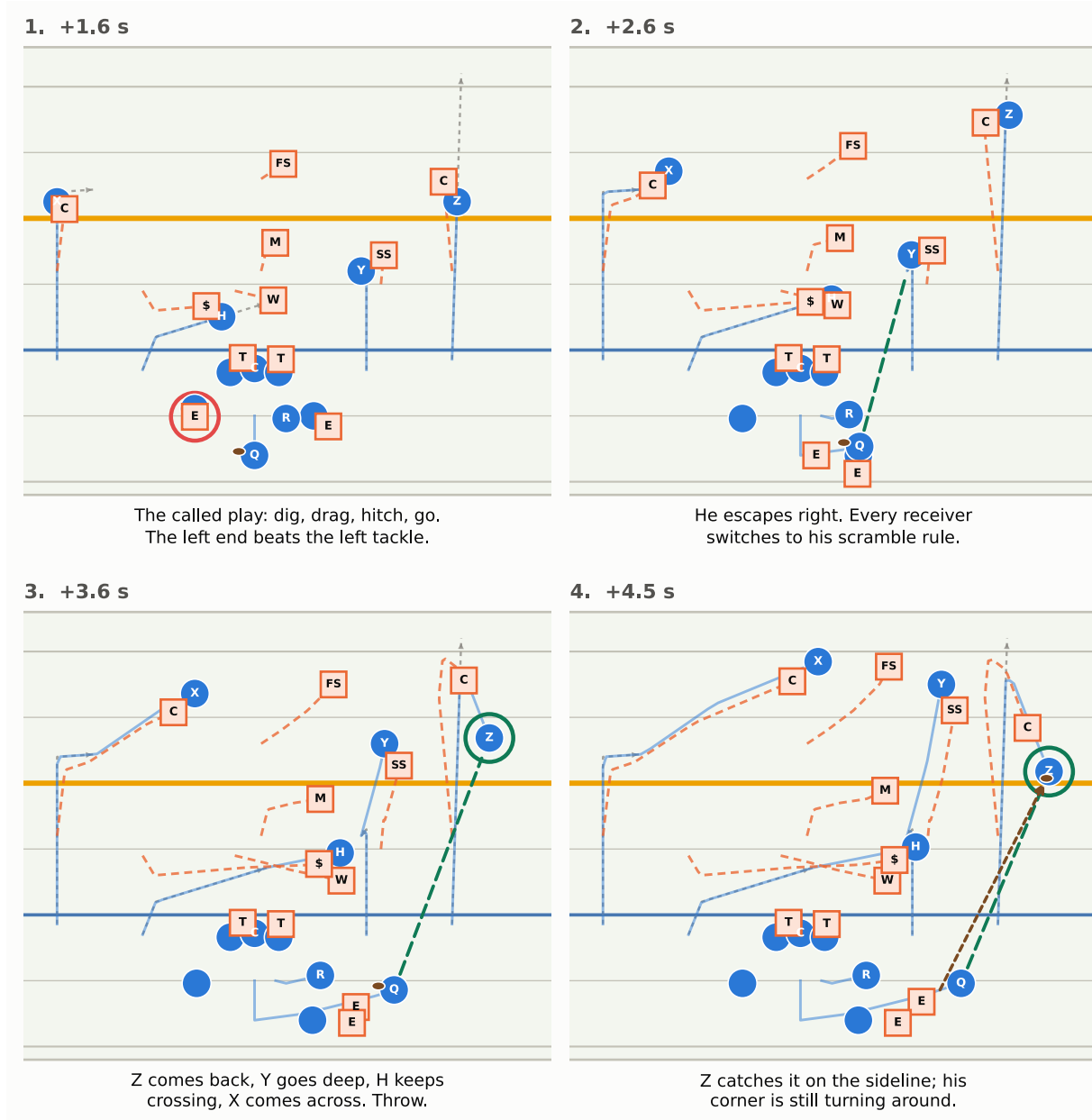


Figure 7: The scramble drill in motion (simulated). The call is a dig by X, a shallow cross by H, a hitch by Y and a go by Z, against man coverage with a free safety deep and a linebacker in the middle; the defense rushes four. Panel 1: the left end beats the left tackle. Panel 2: the quarterback escapes to his right and the receivers switch to their scramble rules. Panel 3: Z has stopped his go route and is working back down the sideline, Y has turned up the field, H keeps crossing, X comes back across; the quarterback throws on the run. Panel 4: Z catches it near the sideline. His cornerback, who was running deep with his back to the quarterback, is still turning around.

! Why it exists

What the scramble drill solves. A pass play is designed to finish in under three seconds. When it doesn't, eleven offensive players are suddenly without instructions, and without rules each receiver would do something different: some keep running their routes into the sideline, some stop, two drift into the same spot. The scramble drill turns the broken play into a second, pre-planned play, so the quarterback's improvisation lands on receivers who are where he expects them.

What it gives up. Nothing in the design, but plenty in the result: a play that goes off script is, on average, a worse play. It is a rescue, not a strategy, and the offense would usually rather the first play had worked. The exceptions are a handful of quarterbacks for whom the rescue is a weapon, and you can find them in the data.

Off-script value: who gains by leaving the pocket?

FTN charts whether the quarterback was outside the pocket on each dropback, whether he left on purpose (a designed rollout) or by escaping. From 2022 to 2025 that happened on 18% of NFL dropbacks. On average those plays were worth +0.011 EPA per play, against +0.049 in the pocket: for the league as a whole, leaving the pocket costs value. For a few quarterbacks it does the opposite.

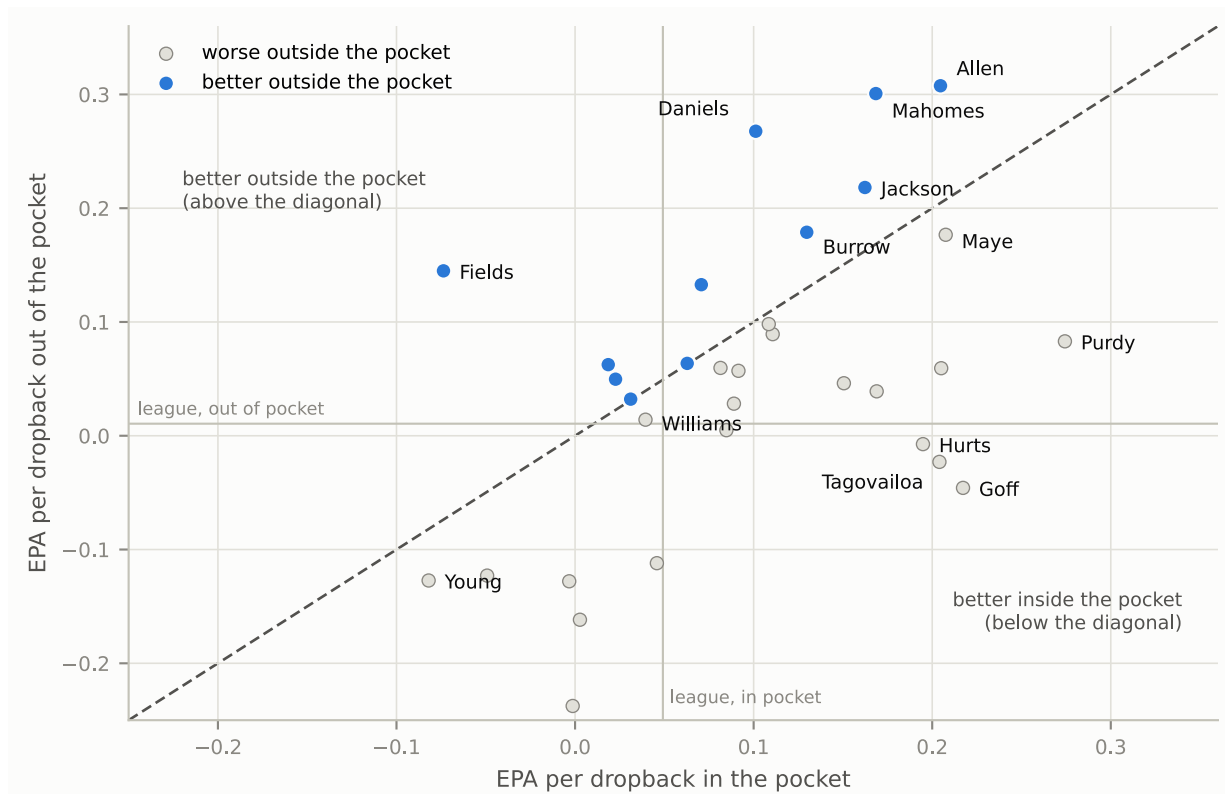


Figure 8: In the pocket versus out of it: every quarterback with at least 800 dropbacks and 150 out-of-pocket dropbacks in the 2022-2025 regular seasons. Horizontal: EPA per dropback (sacks and scrambles included) while in the pocket; vertical: EPA per dropback once outside it, by escape or by design. Points above the dashed diagonal were better outside the pocket than in it, which few quarterbacks were. Josh Allen and Patrick Mahomes sit highest: their off-script plays were worth more than most quarterbacks' on-script ones. Data: FTN charting and nflverse play-by-play via nflreadpy.

Only 11 of the 32 quarterbacks in Figure 8 were better outside the pocket than in it. At the top are the two players the sport thinks of first when it thinks of off-script football: Josh Allen (+0.31 EPA per dropback outside the pocket, +0.20 in it) and Patrick Mahomes (+0.30 and +0.17). Notice what kind of skill that is. Both can run, but the value isn't mainly in the running: it is in throwing accurately on the move, to receivers who know the scramble drill, while the defense is scrambling too.

Read the chart with two cautions. First, "out of the pocket" includes designed rollouts and bootlegs (Play-Action, Bootlegs and Screens), which are good plays for everyone, so a quarterback whose offense calls many of them looks better out there for reasons that have nothing to do with improvisation. Second, it is four seasons of a few hundred plays per quarterback: the order in the middle of the pack is mostly noise, and only the ends of the chart are worth arguing about.

A short history: from Tarkenton to Mahomes

The scramble entered the NFL's vocabulary with Fran Tarkenton, a preacher's son from Georgia who quarterbacked the expansion Minnesota Vikings from their first game in 1961. Behind a thin, new offensive line, he ran around behind it, often for long stretches, until someone came open. Coaches of his era, his own first head coach Norm Van Brocklin among them, thought scrambling undisciplined; Tarkenton's answer, preserved on his Hall of Fame page, was that "I scramble because I'm good at it, because I can twist and dodge those big pass rushers better than most guys, and we get a lot of touchdowns that way." He retired in 1978 holding the career records for completions, passing yards and touchdown passes.³

For the next forty years the pocket passer was the ideal and the scrambler the exception: the quarterbacks who won the most, from Joe Montana to Peyton Manning and Tom Brady, did their damage on schedule, and the offenses built around them, timed to the drop, treated a scramble as a play that had failed. The change in the last decade has two sources. College football filled with spread offenses and athletic quarterbacks who had been improvising since high school, and NFL offenses learned to *coach* the improvisation, practicing the scramble drill weekly instead of hoping for the best. The quarterbacks who came out of that world (Mahomes, Allen, Lamar Jackson, Jalen Hurts) made off-script play part of the plan rather than an emergency.

Getting rid of it: the throwaway

The last of the quarterback's options when nothing is open is the least glamorous and among the most valuable. A **throwaway** is a pass he deliberately throws where nobody can catch it, usually out of bounds or well over a covered receiver's head, to avoid a sack or an interception. He gives up the down but keeps the yards. NFL quarterbacks threw the ball away on 3.5% of dropbacks from 2022 to 2025.

Why is that a skill and not a surrender? Look at what the alternatives cost. In the same seasons, the average throwaway was worth -0.78 EPA (an incompleting, a lost down), the average sack -1.77 (a lost down *and* six or seven yards), and the average interception -4.42. A quarterback who turns two sacks a game into throwaways gains about 2 points a game for his team, without completing a pass.

There is a catch, and it is a rule. **Intentional grounding** is a foul for throwing the ball away to avoid a sack when the pass isn't in the direction and vicinity of an eligible receiver; it costs a loss of down and ten yards (or the spot of the foul, if that is worse), and a safety if the quarterback throws it from his own end zone. The rule changes once the quarterback leaves the **tackle box**, the area between where the two offensive tackles lined up: outside it, any pass that lands at or beyond the line of scrimmage (including out of bounds past it) is legal, receiver or no receiver.⁴ Officiating, Replay, and the Rules of the Ball covers the full rule; here is what it means for the quarterback.

³Pro Football Hall of Fame, "Fran Tarkenton" (career 1961–1978 with the Vikings and Giants; four touchdown passes and a touchdown run off the bench in the Vikings' first game, a 37–13 win over Chicago; records at retirement; the quoted line on scrambling): <https://www.profootballhof.com/players/fran-tarkenton/>.

⁴NFL Rulebook, Rule 8, Section 2 (intentional grounding): a foul if a passer facing an imminent loss of yardage throws a pass not in the direction and vicinity of an originally eligible receiver; not called when a passer who is or has been outside the pocket area throws a pass that lands at or beyond the line of scrimmage extended, including out of bounds. NFL Football Operations, "Intentional Grounding," NFL Video Rulebook: <https://operations.nfl.com/the-rules/nfl-video-rulebook/intentional-grounding/>.

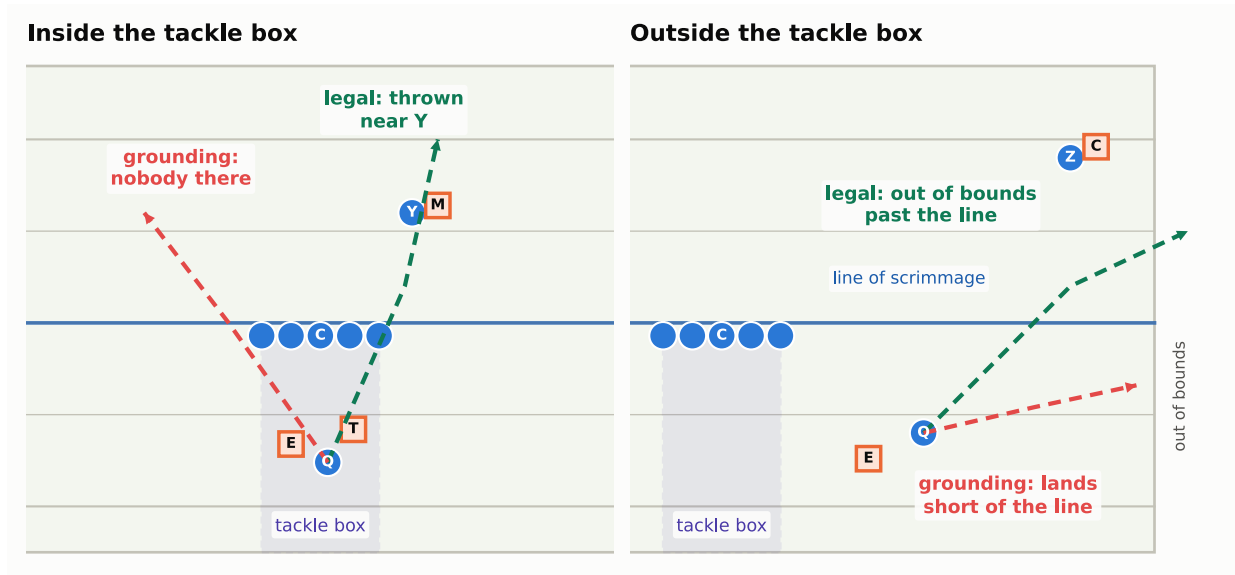


Figure 9: Where a throwaway is legal. Left: inside the tackle box (shaded), a quarterback under pressure must throw in the direction and vicinity of an eligible receiver; throwing it over Y’s head, near him, is a legal throwaway (green), while throwing it to an empty part of the field is intentional grounding (red). Right: once he is outside the tackle box, the ball only has to land at or beyond the line of scrimmage, so throwing it out of bounds past the line is legal (green), but a throw that lands short of the line, near nobody, is still grounding (red). The ball is on the right hash, so the right sideline is close.

The rule rewards exactly the skills this chapter has been describing. A quarterback who climbs and slides inside the pocket can always find a covered receiver to throw near; a quarterback who escapes the pocket earns the right to throw it into the stands. The quarterback who gets in trouble is the one who stays in the box with nobody near him, panics, and flings it at the turf, or who, outside the box, tries to throw it away on the run and doesn’t get it back to the line.

The most famous grounding call of the century shows how expensive the panic is. On the New England Patriots’ first offensive play of Super Bowl XLVI, against the New York Giants in February 2012, Tom Brady dropped back from his own 6-yard line with Justin Tuck bearing down on him, and threw deep down the middle to nobody. The flag came out; Brady had thrown it from his own end zone, so the grounding penalty was a safety, and the Giants led 2-0 before they had run an offensive play. They won by four.⁵

The checkdown: risk management, not cowardice

The last read in most progressions is the **checkdown**: a short, safe throw, usually to a running back or tight end a few yards past the line, underneath all the coverage. It is the quarterback giving up on the big play to take a small, nearly certain one. Fans boo it. Coaches mostly don’t, and the data says why.

⁵“Super Bowl XLVI,” Wikipedia (first Patriots offensive play from their own 6; Brady flagged for intentional grounding in the end zone under pressure from Justin Tuck, a safety; Giants won 21–17): https://en.wikipedia.org/wiki/Super_Bowl_XLVI.

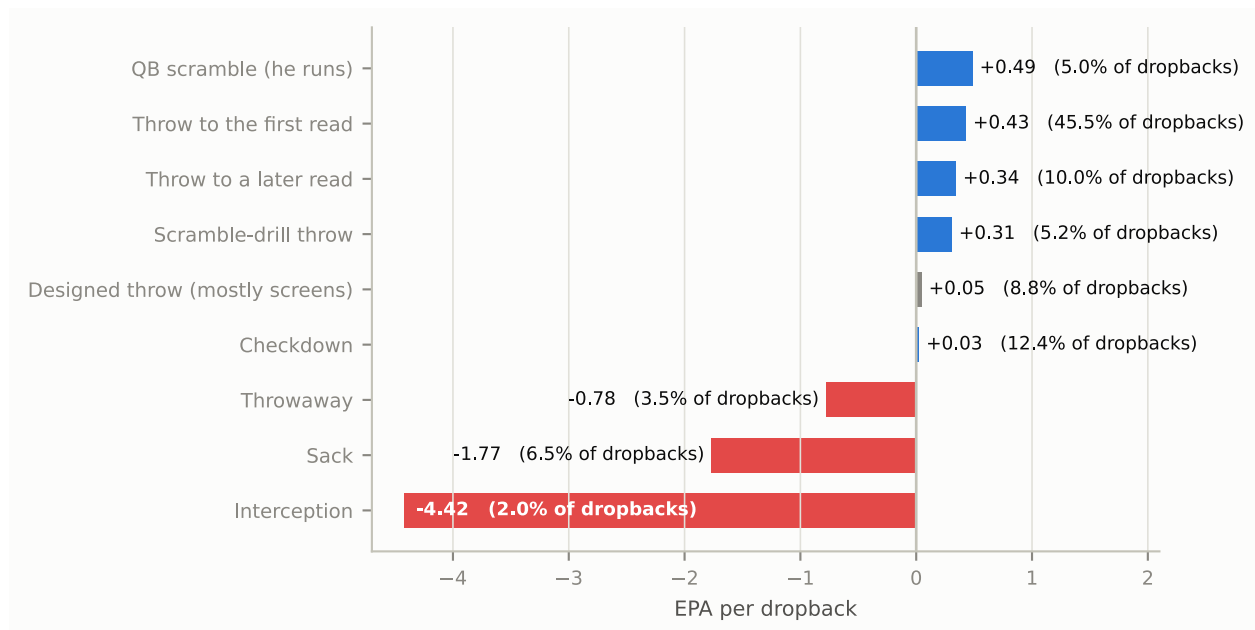


Figure 10: How dropbacks end, and what each ending is worth: every regular-season dropback from 2022 to 2025, grouped by FTN’s charting of where the ball went (best average outcome at the top, worst at the bottom; labels give the share of all dropbacks). A quarterback scramble tops the list because quarterbacks mostly run when the coverage has left open grass. Among throws, the first read is worth the most and each step down the progression gives up value; the checkdown is worth about nothing on average, which is a great deal better than the three endings below it. A throwaway costs a down, a sack a down and yards, an interception the ball. Interceptions and throwaways are counted on their own, whichever read they were aimed at, so the read rows are completions and incompletions only. Data: FTN charting and nflverse play-by-play via nflreadpy.

Read Figure 10 from the middle out. A checkdown is worth, on average, +0.03 EPA: essentially nothing. It is caught 82% of the time, thrown about 0.6 yards past the line, and gains most of what it gains after the catch (6.3 yards on average). That looks like a wasted play only if you compare it to the first read. The real comparison is to what happens to a quarterback who *doesn’t* check the ball down when the first reads are covered: he holds it, and the next three bars are waiting. The checkdown is the cheapest way to end a play that has failed.

That is what *risk management* means at quarterback. Each read in the progression has an expected value and a risk of disaster, and the checkdown is the read with the least of both. A quarterback who takes it at the right moment gives up a little upside to cut off the tail of sacks and interceptions, and over a season that tail decides games.

The right moment depends on the situation, and the third-down numbers show how much. On third down with 7 or more yards to go, a checkdown converted 24% of the time, against 43% for a throw to the first read; on third-and-3 or shorter, a checkdown converted 58%. On third-and-long, then, a checkdown usually means a punt. Even so it averaged -0.20 EPA on those plays, against -1.79 for a sack: when the ball is going to be punted anyway, the yards the back gains after the catch still flip a little field position, while a sack can push a field goal out of range. Good quarterbacks know the down and distance on every snap and adjust: checkdowns early, when four yards on first down keeps the offense on schedule, patience on third-and-long, when only a throw past the sticks will do.

🔥 Common misconception

“Checking it down is what scared quarterbacks do.” Every great quarterback checks the ball down, often, and good offenses design their checkdowns to gain yards: a back who catches it in space with one linebacker to beat. The difference between a good checkdown and a bad one is *when*: a checkdown after the deeper reads were honestly covered is a good decision; a checkdown on the first beat of a play, before the deep reads had time to open, is a quarterback who has stopped trusting his protection. Watch where his eyes were before he threw it.

Measuring a quarterback: time to throw, aDOT, CPOE and EPA

You can now describe a quarterback’s process. The data can describe his *style* and his *results*, and four numbers do most of the work.

- **Time to throw** (seconds from the snap to the release) says how long he holds the ball: quick and decisive, or patient and hunting.
- **aDOT** (average depth of target) says how far downfield he aims: underneath and safe, or deep and aggressive.
- **CPOE** says how accurate he is, after allowing for how hard his throws were.
- **EPA per dropback** says what it all adds up to: the points his dropbacks added, sacks, scrambles and turnovers included.

The first two describe style, and neither is good or bad on its own; the last two describe results. The one that needs explaining is CPOE.

CPOE: completion percentage, corrected for difficulty

Completion percentage is the oldest quarterback number and one of the most misleading, because it rewards easy throws. A quarterback who checks the ball down all day completes 75% of his passes; one who throws deep outs and posts completes 60% and may be the better passer. **CPOE**, or **completion percentage over expected**, fixes this by comparing each throw with a model's estimate of how often a throw like it is completed. The nflverse model behind the numbers here knows each throw's depth, which part of the field it went to, the down and distance, and whether the quarterback was hit as he threw;⁶ the Next Gen Stats version also knows, from tracking data, how far the receiver was from the nearest defender.⁷ A deep out to the sideline might have an expected completion rate of 45%; a checkdown to the flat, 85%. Complete the deep out and you gain 55 percentage points over expected; drop the checkdown and you lose 85. Average those differences over a season and you have CPOE.

⁶Ben Baldwin, "nflfastR EP, WP, and CP models," Open Source Football (Sept 2020), describing the completion-probability model behind the `cp` and `cpoe` columns: <https://www.opensourcefootball.com/posts/2020-09-28-nflfastR-ep-wp-and-cp-models/>.

⁷NFL Next Gen Stats glossary, "Completion Probability" and "Completion Percentage Above Expectation" (built from tracking data, including receiver separation): <https://nextgenstats.nfl.com/glossary>.

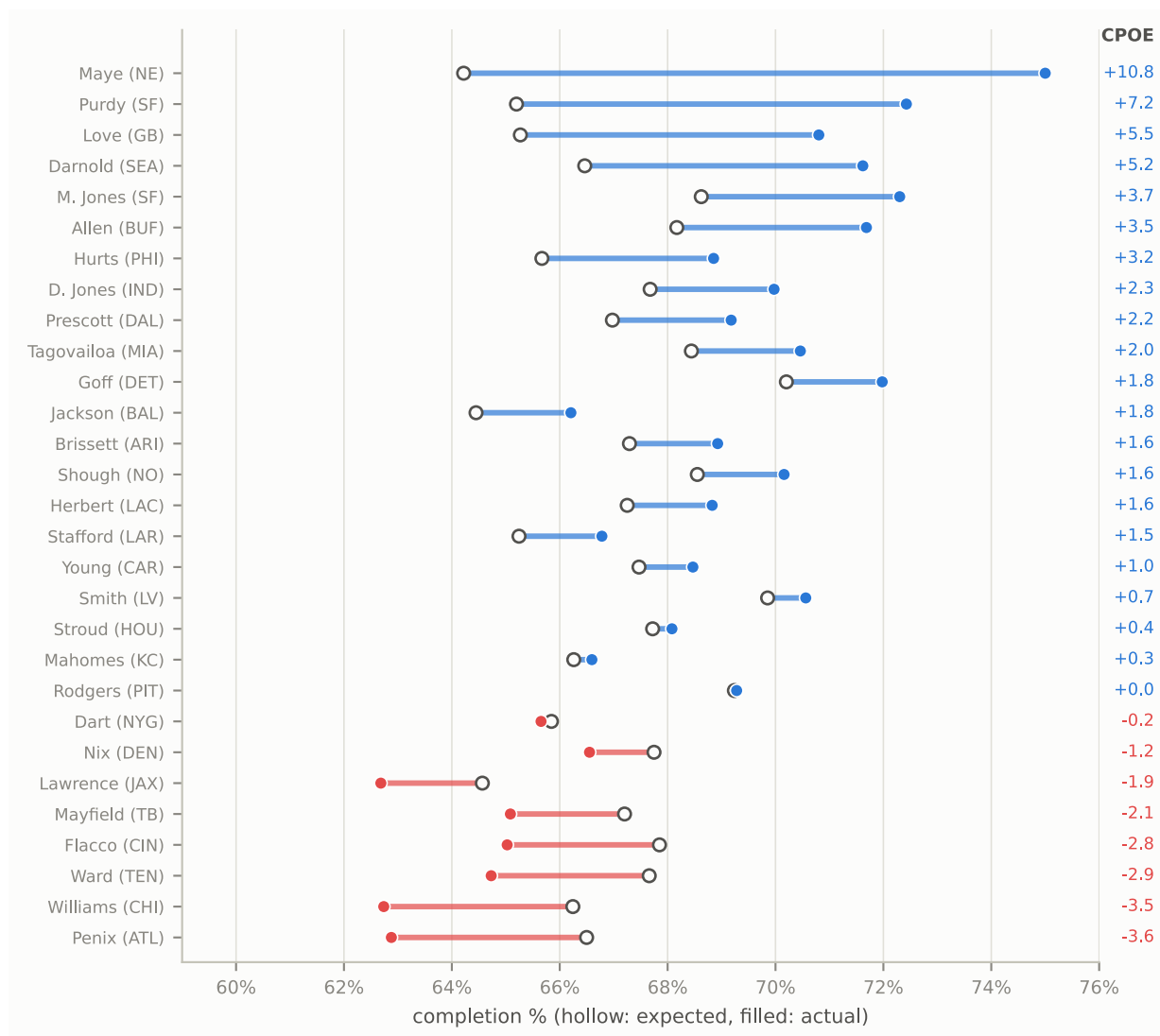


Figure 11: Completion percentage and what was expected, 2025 regular season (quarterbacks with at least 300 dropbacks). Each row is a quarterback: the hollow circle is the completion rate the nflverse model expected from his throws, the filled circle is what he actually completed, and the line between them is his CPOE, the number on the right. Rows are sorted by CPOE. Two quarterbacks with the same completion percentage can be far apart once difficulty is counted. Data: nflverse play-by-play (cp and complete_pass) via nflreadypy.

In 2025 the spread was enormous: Drake Maye completed +10.8 percentage points over expectation, and Michael Penix -3.6. CPOE is the best single public measure of accuracy, and it is steadier from season to season than completion percentage, but read it with two cautions. It credits the quarterback for everything that is not in the model: a receiver who makes a great catch, or a scheme that gets receivers wide open, raises it. And it says nothing about *where* the ball went, only whether it was caught. A quarterback who throws safely and accurately to covered checkdowns can post a fine CPOE and a poor offense.

🔥 Common misconception

“Completion percentage measures accuracy.” It measures accuracy *and* the difficulty of the throws a quarterback chooses, mixed together, and difficulty dominates. A conservative passer and an aggressive one can be equally accurate and be ten points apart in completion percentage. If you want accuracy, look at CPOE; if you want to know how aggressive he is, look at aDOT; if you want to know whether it worked, look at EPA.

The style map

Put time to throw and aDOT on two axes and color each quarterback by his EPA per dropback, and quarterback styles separate into families.

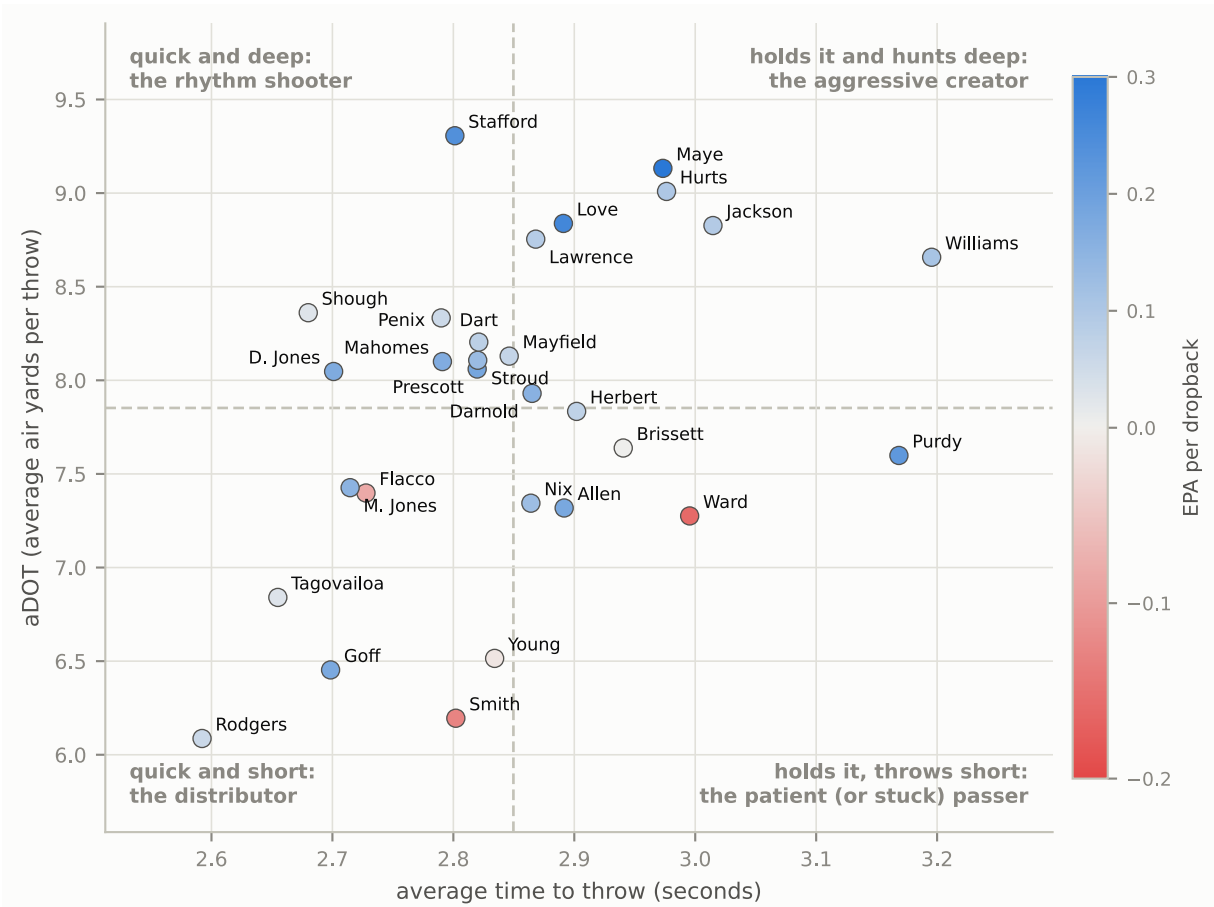


Figure 12: Quarterback style map, 2025 regular season: every quarterback with at least 300 dropbacks. Horizontal: average time to throw (Next Gen Stats, snap to release, sacks excluded). Vertical: aDOT, the average air yards of his throws. Color: EPA per dropback (sacks, scrambles and turnovers included), blue above zero and red below. The dashed lines are the averages of the quarterbacks shown, splitting the map into four styles. Style is not success: there are blue and red dots in every corner. Data: nflverse play-by-play and Next Gen Stats via nflreadpy.

The four corners of Figure 12 are four ways to play the position.

- **The distributor** (quick and short) wins with decisions: the ball comes out on rhythm to the first open receiver, and the offense lives on YAC. He rarely takes sacks, and his ceiling depends on how many yards his receivers make after the catch.
- **The rhythm shooter** (quick and deep) is the rarest type, because deep routes take time to develop; it takes anticipation and a scheme that gets receivers vertical fast.
- **The aggressive creator** (holds it, throws deep) extends plays to give deep routes time, with a pocket he manages and an off-script game he trusts. His sack rate is higher and his big-play rate too.
- **The patient or stuck passer** (holds it, throws short) is the corner to worry about: the ball comes out late *and* short, which usually means the first reads were not open and the quarterback ended up checking it down after a long look. Sometimes that is a scheme or a line problem rather than a quarterback problem, but it is rarely a good sign.

In 2025, the league MVP, Matthew Stafford, sat in the top half of the map: an aDOT of 9.3 yards at 2.80 seconds, and +0.24 EPA per dropback. Drake Maye, the runner-up, threw just as deep with the league's best CPOE and its best EPA per dropback (+0.33). Brock Purdy sits deep in the bottom-right corner, holding the ball longer than almost anyone, and still posted +0.22: the patient passer, not the stuck one. Style tells you how a quarterback plays; the color tells you whether it worked, and every corner has both colors in it.

Why the quarterback dominates value

One last number explains why everything in this chapter matters so much, and why the quarterback is paid like no one else on the field.

First, he touches the ball on almost every play, and on most of them he decides where it goes: in 2025, 60% of NFL offensive plays were dropbacks. Second, passing plays vary more than running plays, and the quarterback is the player who decides which way they go. Among 2025 starters with 300 or more dropbacks, EPA per dropback had a standard deviation of 0.110; among ball carriers with 150 or more carries, EPA per carry had one of 0.075. Multiply by a season's workload (536 dropbacks for the median qualifying quarterback, 232 carries for the median back) and a quarterback one standard deviation above average is worth about 59 points over a season, against about 17 for a back.⁸

Both numbers share credit with the blockers, the receivers and the play-caller, and the split is unknowable from the box score. But the gap is too big to be an accident. The quarterback is the player with the most chances to add or lose points, on the plays where the most points are at stake. That is the starting point of Building a Roster, where the salary cap turns this gap into the most important economic fact in the sport.

Film room

i Film room: Tom Brady and Peyton Manning — the pocket as a craft

Peyton Manning, Denver Broncos, 2013. Manning set the single-season records for touchdown passes (55) and passing yards (5,477) in 2013, at 37.⁹ The number that explains how is a quieter one: he was sacked on 2.7% of his dropbacks, the lowest rate of any regular starter that season (the league rate was 6.4%).¹⁰ Manning had famously little interest in running. His pocket skill was making the decision before the rush arrived: an elaborate pre-snap read (The Pre-Snap Phase) that told him which side to work, then a progression run so fast that the ball was usually gone before pressure mattered.

Tom Brady, 2000-2022. Watch any Brady dropback and look at his feet, not his arm. When the edge closed he moved up into the pocket a step or two, rarely more, and kept his eyes downfield; when an interior rusher came he slid a yard and reset. He did it so economically that it looks like nothing happened, which is the point: a quarterback who climbs well never seems to be under pressure at all. He also got rid of the ball. Even at 44 and 45 his average time to throw was among the quickest in the league: 2.50 seconds in 2021 (rank 2 of 30 qualifying passers) and 2.45 in 2022 (rank 1 of 30).¹¹

What to notice: neither man was a scrambler, and both were extraordinarily hard to sack. Pocket skill is mostly decision speed and footwork, not escapability.

i Film room: Kansas City Chiefs 2018 — Mahomes's left hand

Monday night, October 1, 2018, at Denver. The Chiefs trailed 23-20 with 3:14 left and faced third-and-5. Patrick Mahomes, in his first season as a starter, was flushed to his left, and Von Miller was dragging him down. On the way to the ground he switched the ball to his left hand and flipped it to Tyreek Hill for six yards and a first down. Kansas City scored the winning touchdown seven plays later. Asked later for Mahomes's greatest play, Andy Reid chose that one: a short throw, made with the wrong hand, with the game on the line.¹²

What to notice: the play was over by every normal standard: no pocket, a Hall of Fame-level rusher on his back, his body going the wrong way. Off-script value is mostly made of throws like this: short, ugly and converting. Across 2022-2025 Mahomes's dropbacks outside the pocket were worth +0.30 EPA each, more than his dropbacks inside it (Figure 8).

⁸ Author's calculation, nflverse play-by-play, 2025 regular season: EPA per dropback (sacks and scrambles included) for passers with at least 300 dropbacks; EPA per carry on non-dropback runs for ball carriers with at least 150 carries. A rough comparison of spreads, not a measure of individual credit; see Expected Points, EPA, and Success Rate.

⁹ "2013 Denver Broncos season," Wikipedia (Manning's 55 touchdown passes and 5,477 passing yards, both single-season records at the time): https://en.wikipedia.org/wiki/2013_Denver_Broncos_season.

¹⁰ Author's calculation, nflverse play-by-play, 2013 regular season: sacks per dropback, passers with at least 300 dropbacks.

¹¹ Next Gen Stats average time to throw (snap to release, sacks excluded), regular season, passers with at least 300 attempts, via nflverse/nflreadpy ([load_nextgen_stats](#)).

¹² CBS Sports, "Patrick Mahomes' greatest play was this short throw, according to Chiefs coach Andy Reid" (third-and-5, 3:14 left, trailing 23-20; six-yard completion to Tyreek Hill while being dragged down by Von Miller; winning touchdown seven plays later): <https://www.cbssports.com/nfl/news/patrick-mahomes-greatest-play-was-this-short-throw-according-to-chiefs-coach-andy-reid>. Final score 27-23: ESPN, https://africa.espn.com/nfl/story/_/id/24866002/patrick-mahomes-sparks-chiefs-comeback-win-broncos.

i Film room: New York Giants 2007 season, Super Bowl XLII — the escape

February 3, 2008. The Giants trailed the unbeaten Patriots 14-10 and faced third-and-5 at their own 44 with 1:15 left. Eli Manning took the snap in the shotgun and the pocket collapsed: Jarvis Green and Richard Seymour got their hands on his jersey and should have had the sack. Manning spun free, backed out of the pile, and threw high down the middle to David Tyree, who had worked back into the middle of the field when his route was disrupted. Tyree caught it against his helmet with Rodney Harrison draped over him, at the Patriots' 24. The Giants scored four plays later and won 17-14.¹³

What to notice: escape and scramble drill together. Manning was no runner; he simply refused to go down and kept his eyes downfield while he got loose. Tyree did what the drill tells a receiver to do when the play breaks: get to open grass in the quarterback's line of sight, on a different level from everyone else.

i Film room: Buffalo Bills 2024 — Josh Allen, MVP off script

Josh Allen won the 2024 MVP award with a season that mixed both halves of this chapter.¹⁴ Across 2022-2025 his dropbacks outside the pocket were worth +0.31 EPA each, the highest of any quarterback with a real sample, and higher than his own dropbacks inside it (+0.20). Allen is enormous (listed at 6-foot-5 and about 237 pounds), fast, and has one of the strongest arms in the league, which together make the scramble drill unusually dangerous: he can throw 50 yards on the run, so the defense's deep defenders cannot come up to chase him, and he can run through a tackle, so the underneath defenders cannot drop off to cover.

What to notice: Allen's early career was full of off-script plays that went wrong (forced throws across his body, sacks taken while looking for something big). The MVP version still improvises, but the throwaway and the checkdown are in his game too. Off-script talent becomes off-script value when it comes with the judgment to give up on a play.

i Film room: Cincinnati Bengals 2024 — Joe Burrow's timing

Joe Burrow led the NFL in passing yards (4,918) and touchdown passes (43) in 2024.¹⁵ He did it in the pocket. Burrow is not an escape artist; his gifts are the ones in the first half of this chapter: anticipation (throwing outs and digs before the receiver breaks, especially to Ja'Marr Chase, whose timing he has known since they were college teammates at LSU) and subtle movement in the pocket (a step up, a half-slide) that buys the extra beat for the throw.

What to notice: on replays of Bengals completions, freeze the frame at the release and find the receiver. Very often he hasn't turned around yet.

¹³"Helmet Catch," Wikipedia (third-and-5 at the Giants' 44 with 1:15 left; Seymour and Green nearly sacked Manning; 32-yard completion to the Patriots' 25; Tyree pinned the ball to his helmet with Rodney Harrison defending): https://en.wikipedia.org/wiki/Helmet_Catch.

¹⁴NFL.com, "Bills QB Josh Allen wins 2024 AP NFL Most Valuable Player award": <https://www.nfl.com/news/bills-qb-josh-allen-wins-2024-ap-nfl-most-valuable-player-award>.

¹⁵Passing yards and touchdown passes from nflverse play-by-play, 2024 regular season (author's calculation via nflreadpy; Burrow ranked first in both).

i Film room: San Francisco 49ers 2023 — Brock Purdy and the system-quarterback debate

Brock Purdy was the last player chosen in the 2022 draft, number 262, the pick nicknamed “Mr. Irrelevant.”¹⁶ In 2023, his first full season, he led the 33 quarterbacks with 250 or more dropbacks in EPA per dropback (+0.32, rank 1) and in the nflverse model’s CPOE (rank 1). The argument about him since has been whether those numbers are his or Kyle Shanahan’s: a scheme famous for motion, play-action and receivers who run after the catch, with Christian McCaffrey, Deebo Samuel, Brandon Aiyuk and George Kittle to throw to. Is he a **system quarterback**?

The data cuts both ways, which is why the argument never ends. Against the “YAC merchant” version of the charge: in 2023, 48% of his receiving yards came after the catch, about the league rate (48%), and his aDOT (8.2) was above the league average (7.8). He was throwing downfield and completing it. For the charge: Next Gen Stats’ completion model, which knows how open each receiver was, ranked his completion percentage over expected only 5th that season, well below the nflverse model’s ranking; some of his accuracy was receivers being open, which the scheme creates. And in 2025, when injuries cost Purdy much of the season, his backup Mac Jones ran the same offense at +0.15 EPA per dropback, against Purdy’s +0.22: worse, but far better than Jones’s −0.06 across 2022–2024 with New England and Jacksonville.

What to notice: the question “is it the quarterback or the system?” is the right one to ask, and CPOE models that know about separation help answer it. But a quarterback who throws on time to the right read is part of what makes a system work. Watch Purdy’s anticipation on intermediate throws and judge for yourself.

Watch for it on Sunday

💡 Watch for it

Watch the eyes. On replays, especially the end-zone camera that looks straight at the quarterback, count how many reads he got through before he threw.

- **Find his first look.** Where do his eyes (and his helmet’s facemask) point as his drop ends? That is usually read 1. Did the ball go there? More than half the time it does.
- **Count the head turns and hitches.** Each snap of the head to a new area, usually with a hitch step, is a new read. One look and a throw: a rhythm throw. Two or three: a progression. A long stare at one side: he was waiting for something to open, or he was confused.
- **Note the time.** Count “one-Mississippi” from the snap. Ball out before two: quick game or first read. Out after three: something went wrong, or the play was a deep shot with max protection.
- **Watch his feet when the edge closes.** Step up (climb)? Slide? Or drift back into the rush? The good ones move forward.
- **When he leaves the pocket, watch the receivers.** Did the deep ones come back and the short ones go deep? Is everyone moving toward his side? That is the scramble drill. Who got open?
- **Judge the ending.** A checkdown on third-and-12 is a punt with a few extra yards; on first-and-10 it is a fine play. A throwaway when nothing is there is a small win. A sack after four seconds is usually on the quarterback.

For your prediction log: for one quarter, write down for each dropback how many reads the quarterback got through (1, 2, 3, or “off script”), and whether the ball came out on time. By the end you will know which kind of quarterback you are watching.

¹⁶NFL.com, “Mr. Irrelevant: 49ers select Iowa St. QB Brock Purdy to end 2022 NFL Draft”: <https://www.nfl.com/news/mr-irrelevant-49ers-select-iowa-st-qb-brock-purdy-to-end-2022-nfl-draft>. The 2023 and 2025 numbers are the author’s calculations from nflverse play-by-play and Next Gen Stats.

Predict the play

💡 Predict the play 1: the pocket at two seconds

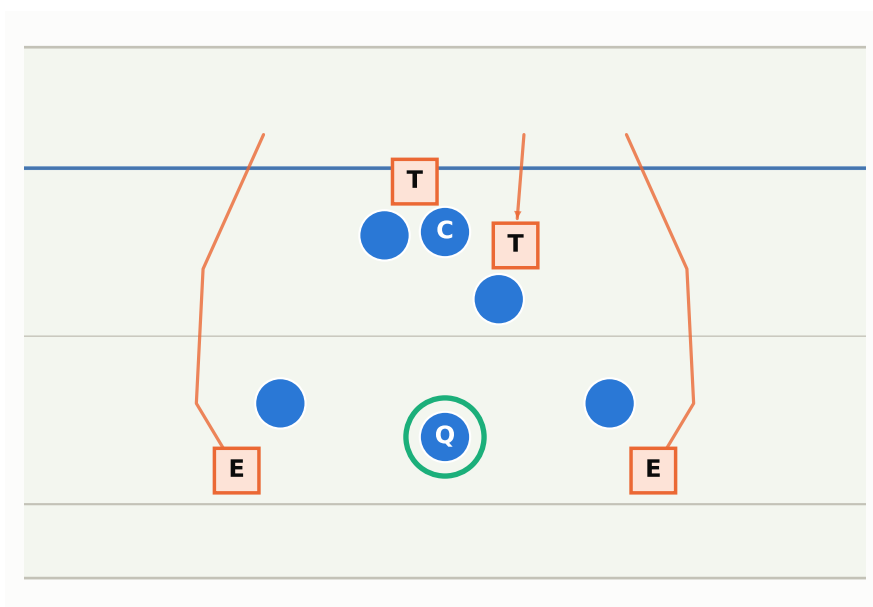


Figure 13: Freeze frame, about two seconds after the snap. Both ends have run their arcs and the tackles have pushed them deep. On the right, the defensive tackle has driven the right guard back about two yards. The left side of the interior is holding. The quarterback (ringed) is at his launch point, eight yards deep.

Where should the quarterback go: back, up, left, right, or out of the pocket?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: third-and-9 at the 31

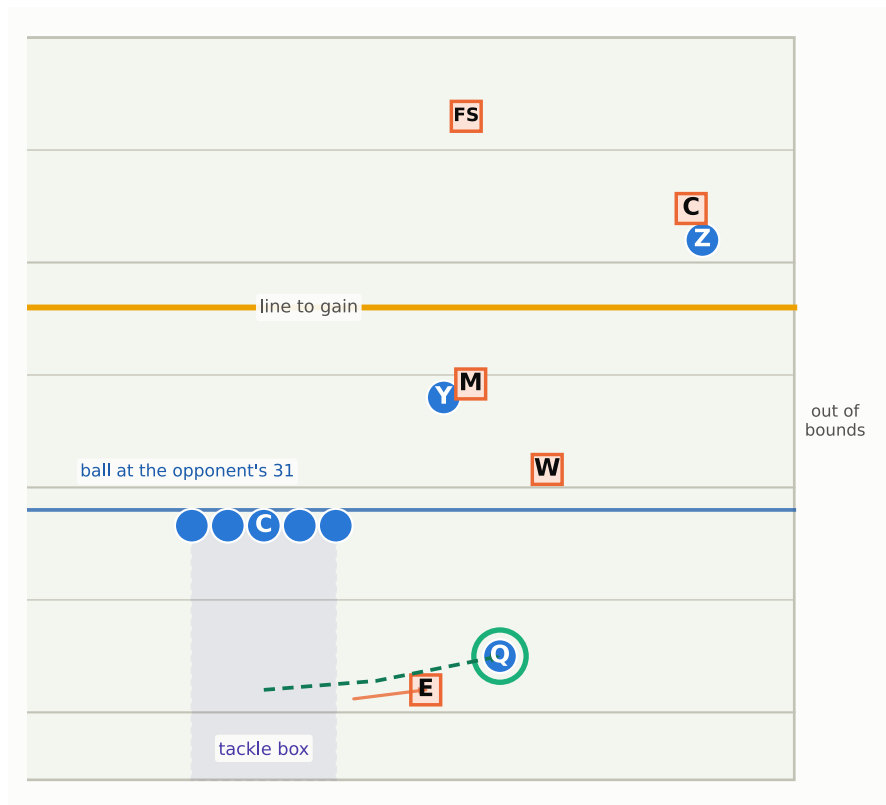


Figure 14: Third-and-9 at the opponent's 31-yard line, so a field goal would be about a 48-yard try. The quarterback has escaped to his right, outside the tackle box (shaded), with the right end chasing him. Z, his only receiver on that side, is running down the sideline with the cornerback on top of him; Y is covered underneath by the Mike. There is no running room to the first-down line: the Will has come up to spy him.

What should the quarterback do with the ball, and is it legal?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: he breaks left

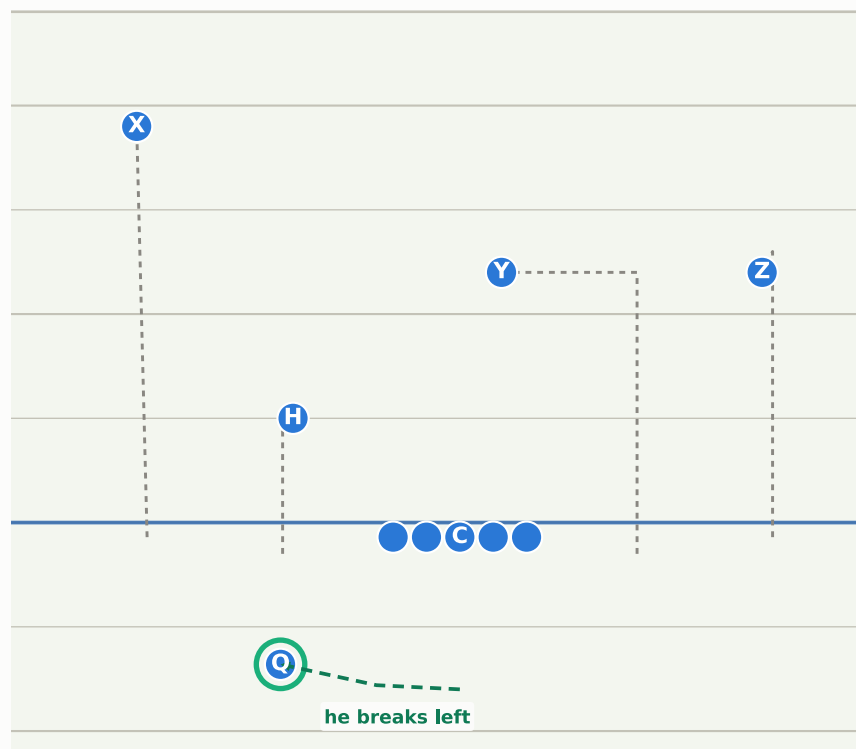


Figure 15: The called play has broken down and the quarterback (ringed) has escaped to his LEFT. The receivers are shown where they are at that moment, with their called routes as grey dashes: X (left outside) was running a go and is 19 yards deep; H (left slot) was on a 5-yard hitch; Y was running a dig from the right that has reached the middle; Z (right outside) was running a 12-yard curl on the far side. The defense is not shown.

Using the scramble-drill rules, where should each of X, H, Y and Z go, and where will the quarterback look first?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **The quarterback's eyes run on his feet's clock.** Protection check at the snap, read 1 on the last step of the drop, a hitch for each later read, then the checkdown. Each later read costs about half a second, and value.
- **He reads defenders, not receivers, and throws before they are open.** Each read has a key defender. An anticipation throw, released before the break, beats the defenders' reaction; a "see it, throw it" ball arrives a beat late into a closing window.
- **Pocket movement is a response to the rush picture.** Edges pushed past: climb. Pressure from one side: slide. Interior collapsed: escape, usually outside the tackle on the side the end was run past. Eyes stay downfield throughout.
- **Off-script football is coached.** The scramble drill (short goes deep, deep comes back, go where he goes, backside comes across deep) turns a broken play into a second play. For most quarterbacks leaving the pocket costs value; for a few (Allen, Mahomes) it adds it.
- **Bad endings have a price list.** A checkdown is worth about nothing, a throwaway a down, a sack a down and yards, an interception the ball. Throwaways must avoid intentional grounding: near a receiver inside the tackle box, past the line of scrimmage outside it.
- **Four numbers profile a passer.** Time to throw and aDOT describe style; CPOE (completions over expected) and EPA per dropback describe results. Because he decides most plays, and passing varies more than running, the quarterback swings more points than anyone else on the field.

Connections

Builds on: Passing Game Fundamentals (the drop as a clock, progressions, aDOT); Pass Protection (the pocket as a cup, the launch point, time to throw and pressure); The Quick Game (rhythm throws and the pre-snap read); and Dropback Concepts (the numbered reads of each concept).

Sets up: Offensive Systems and Play-Calling Languages, where the progression becomes part of a system's language; Concepts Versus Coverages, which teaches how the quarterback reads coverage and moves safeties with a look-off (his eyes pointing one way to move a defender before he throws the other); Disguise, Rotation, and Split-Field Coverage, on defenses built to make the pre-snap read wrong and the post-snap read late; Officiating, Replay, and the Rules of the Ball, for intentional grounding and the tackle box in full; and Building a Roster, on what the quarterback's value means under a salary cap.

Case study: Andy Reid's Chiefs, an offense built to make the most of a quarterback who is as dangerous off script as on it.

Go deeper

- Ron Jaworski with Greg Cosell and David Plaut, *The Games That Changed the Game* (2010): a quarterback's eye view of famous game plans, by a former NFL quarterback and the NFL Films analyst who watched more all-22 film than anyone.
- Chris B. Brown, *The Art of Smart Football* (2015) and the Smart Football archive: short essays on progressions, reads and how quarterbacks are taught to see the field.
- The all-22 "coaches film" on NFL+ Premium, which shows every player on every snap from above: the only way to check a quarterback's progression against what was actually open.
- Ben Baldwin's open-source work on the nflverse CPOE and EPA models (Open Source Football), and the Next Gen Stats passing leaderboards for time to throw, aggressiveness and completion probability.

Answers to Predict the play

Answer 1.

Up, and a little to his left: climb into the space behind the left guard and the center. Read the rush picture piece by piece. The ends have been pushed past him, so drifting back walks him into them, and escaping outside means running around a rusher who is already at his depth. The right side of the interior is caving: the tackle has the right guard two yards deep, so stepping straight up, or up and right, takes him into that collision. The left side of the interior is holding firm, and that is where the cup still has room.

So he climbs at an angle, a step or two up and half a step left, which is a climb and a slide at once. He doesn't have to leave the pocket, his eyes stay downfield, and he is still inside the tackle box, so if nothing is open he can still throw it near a covered receiver or check it down. Escaping would only be right if the left side collapsed too.

Answer 2.

Throw it away, out of bounds past the line of scrimmage, and kick the field goal. It is legal because he is outside the tackle box. Walk through his options. Z is covered with the cornerback on top, so a throw to him is a 50-50 ball at best and an interception at worst, and an interception here throws away a likely three points. Y is covered by the Mike underneath. Running gets nowhere near the line to gain with the Will spying him. The sack is the disaster that's easiest to miss: an eight-yard sack pushes a 48-yard field goal back to about 56 yards, where even good kickers miss often.

So the ball goes out of bounds beyond the line of scrimmage, ideally over Z's head toward the sideline. From inside the tackle box that throw would have to land near an eligible receiver; from where he is, outside the box, it only has to reach the line. A throwaway costs a down, which on third down he was about to lose anyway, and keeps every yard. That is exactly the bargain in Figure 10: a throwaway at about -0.8 EPA against a sack at about -1.8 .

Answer 3.

X comes back down the left sideline; H takes off deep; Y keeps crossing to the left at his depth; Z works back across the field, deep. The quarterback looks deep to short: H or Z first, then X, then Y.

- **X** is deep, so he comes back: he stops his go route and works back toward the line of scrimmage along the left sideline, on the quarterback's side, into the space his cornerback (running deep with him) leaves behind.
- **H** is short, so he goes deep: he turns up the field from his hitch, behind the defenders who have turned to chase the quarterback.

- **Y** is a crosser, and he keeps going where the quarterback goes, toward the left, at about 12 yards, a different level from H above him and X coming back below.
- **Z** is on the backside. Standing on a curl 30 yards from the quarterback is useless, so he runs back across the field, deep, into the middle the defense has emptied, and into the quarterback's line of sight.

That gives four receivers at four depths, all moving toward the quarterback's side. He looks deep first (H, or Z if the middle is empty), then X coming back (a short, easy throw toward the sideline he is running at), then Y. If all four are covered he has two good options left: run, if there is grass to the line to gain, or throw it out of bounds past the line; he is outside the tackle box. What he must not do is throw back across his body to the right.